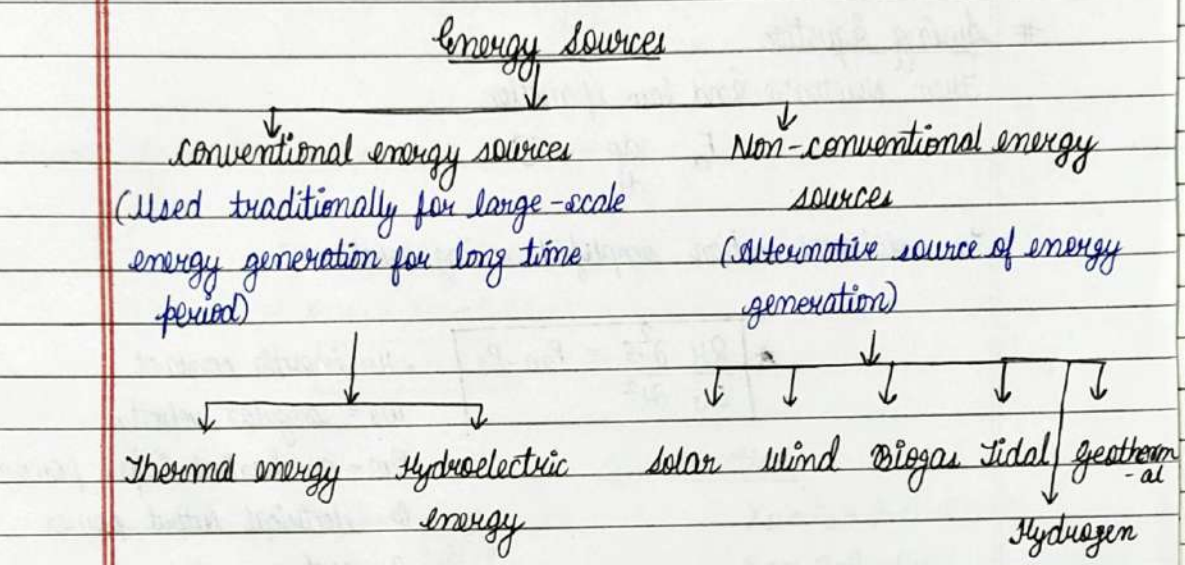
Non-conventional Energy Sources

In an electrical power system, energy is converted into electrical energy from different primary sources.

Example :-

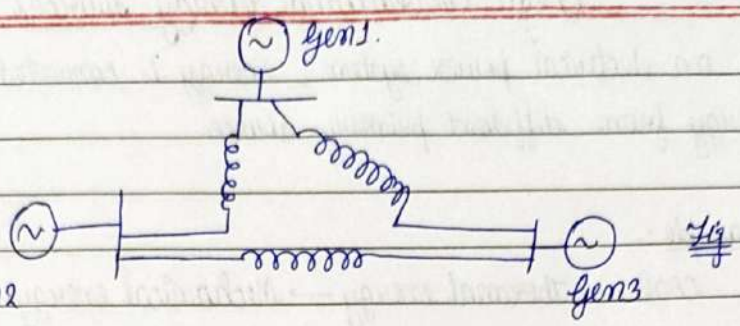
coal → thermal energy → Mechanical energy → Electrical energy



NOTE :-
- India's total installed electricity-generation capacity is about 548.86 GW
- Thermal, hydro and nuclear source accounts for 312.33 GW (56.9% contribution by conventional sources).
- Non-conventional sources accounts for 236.53 GW (about 43.1% of installed capacity).

Multi-generator power system/grid

Different generating stations → transmission network → consumers



Swing equation

From Newton's 2nd law of motion

$$F_{ex} = \frac{dp}{dt} \longrightarrow (1)$$

For rotational motion, simplified swing equation is :-

$$\star \boxed{\frac{2H}{\omega_s}\frac{\partial^2 \delta}{\partial t^2} = P_m - P_e}$$

, H = inertia constant
ω_s = angular velocity
P_m = mechanical input power
P_e = electrical output power
δ = rotor angle

Synchronous generator for thermal power system modelling

$$\frac{d\delta}{dt} = \omega_B S_m$$

and $$\frac{dS_m}{dt} = \frac{1}{2H}(1 - 2.5 \sin\delta)$$

Q) $\omega_B = 377$ rad/s, $H = 5$, $\delta_0 = 50^\circ$, $S_m = 0$.

$\Rightarrow$ at $t=0$, $\frac{dS_m}{dt} = \frac{1}{10}(1 - 2.5 \sin 50^\circ)$

$$\Rightarrow \boxed{\frac{dS_m}{dt} \approx -0.091}$$

Model is using 60-Hz power system (377 rad/s = $2\pi \times 60$).

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Euler's method

$\frac{dx}{dt} = f(x)$ · tells us how fast X is changing

Euler's method says :-

New value = old value + (rate of change)

$$\boxed{X_{n+1} = X_n + \Delta t \left(\frac{dx}{dt}\right)_n}$$

Example: $\frac{dx}{dt} = -2X$ and $X(0) = 5$, $\Delta t = 0.1$

$$X_{n+1} = X_n + 0.1(-2X_n)$$

$$\Rightarrow X_{n+1} = X_n - 0.2X_n$$

$$\Rightarrow \boxed{X_{n+1} = 0.8X_n}$$

Iteration 1 :-

$$X_1 = X_0 + \Delta t \left(\frac{dx}{dt}\right)_0$$

$$\Rightarrow X_1 = 5 + (0.1)(-10)$$

$$\Rightarrow X_1 = 5 - 1$$

$$\boxed{X_1 = 4}$$

$$\therefore \boxed{X(0.1) = 4}$$

Iteration 2 :-

$$X_2 = X_1 + (0.1)(-2X4)$$

$$X_2 = X_1 - 0.8$$

$$X_2 = 4 - 0.8$$

$$\therefore \boxed{X_2 = 3.2} \approx \boxed{X(0.2) = 3.2}$$

Iteration 3 :-

$$X_2 = X_2 + 0.1(-2X3.2)$$

$$X_3 = 3.2 - 0.64$$

$$\boxed{X_3 = 2.56}$$

Iteration 4 :-

$$X_4 = 0.8 X 2.56$$

$$\boxed{X_4 = 2.048}$$

Table

iteration	Δt	X
0	0.0	5.000
1	0.1	4.000
2	0.2	3.200
3	0.3	2.560
4	0.4	2.048

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Question ① Total generation of Indian power grid
(contribution of conventional & non-conventional sources)

② Types of conventional & non-conventional srs.

③ Write the swing eqn for thermal power modelling (2nd order & Python code)?

Solar Energy sources

→ Total solar energy production in India is ____ GW.

→ Installed capacity → 160 GW

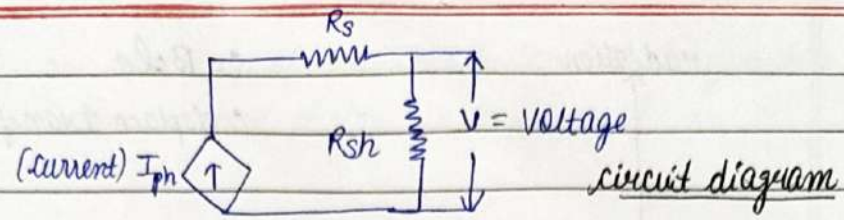
solar cell (Governing equation)

I-V characteristics of a solar cell :- is modeled using the single diode.

$$I = I_{ph} - I_0 \left[\exp\left(\frac{qV + qIR_s}{nKT}\right) - 1 \right] - \frac{V + IR_s}{R_{sh}}$$

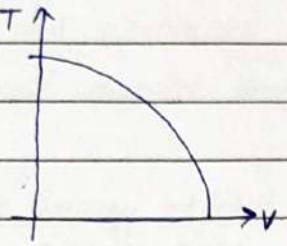
where I = output current
I_{ph} = photocurrent generated by sunlight
I_0 = reverse saturation current of diode
V = terminal voltage
R_s = series resistance
R_{sh} = shunt resistance
q = electron charge (1.602×10^{-19} C)
K = Boltzmann constant (1.38×10^{-23} J/K)
T = cell temperature in kelvin
n = diode ideality factor

Remember, $P = VI$



circuit diagram

- Graph :-



Conclusion :-

$$I_1 \omega_1 = I_2 \omega_2$$

Potential ques

\# MATLAB code for IV-characteristics.

Qy Write C/C++/Python/Java and run it.

Qy Write I-V characteristics of solar cell (write everything — 8 Marks).
(Circuit, Description, graph, code, equation).

Qy Temp / Inadiance (I-V) and (P-V)

\# change the code to draw fig1 and fig2 respectively.

\# Solar energy

Solar radiation is the radiant energy emitted by SUN across an electromagnetic spectrum. It drives earth weather, climate and plant growth.

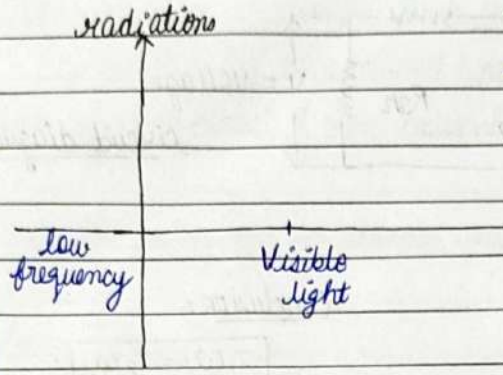
- This energy is measured in watt per square meter (w/m^2),

Using meterological tools like pyranometers and pyrheliometers.

Qy What is solar radiation?

Solar radiation is a stream of energy packet in light waves that flows from the Sun to the earth.

(i) Direct (ii) Diffuse (iii) Global (iv) Spectrum mix
(43 % visible light
49 % infrared
7 % ultraviolet),

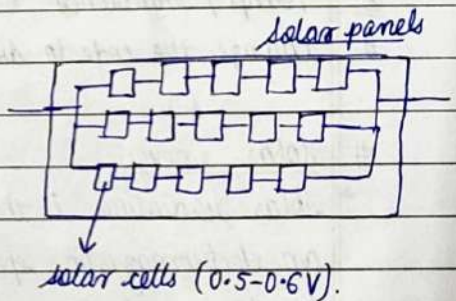


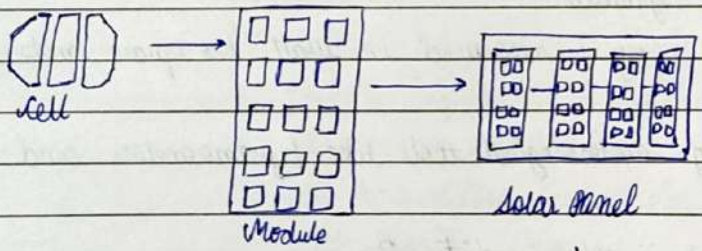
- Bole
- Laplace transform

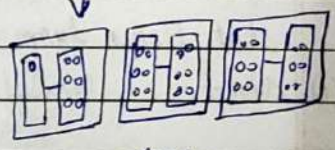
Solar cell or photoelectric cell

Solar cell or photoelectric cell is an electrical device that converts energy of light into electricity by photo voltaic effect which is a physical and chemical phenomenon.
(discovered in 1905 by Edmond Becquerel)

O.C voltage (0.5 - 0.6 Volts). used in solar cells.
(Open-circuit voltage)







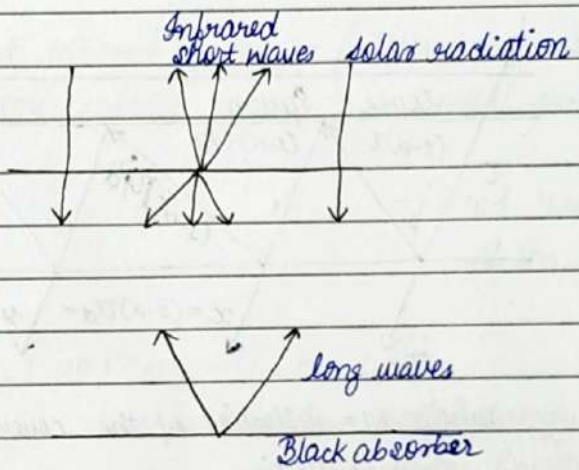
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- Solar cell are made up of semi-conductor which generate free electrons when photons are received.

→ 'Si' is commonly used.
→ Efficiency of 12-15%.
→ 2000 MW Karnataka
→ Kurnool district, Andhra Pradesh (capacity :- 1000 MW (or 1GW))

Solar Energy → Heat (conversion)

In greenhouse, the sunlight enter through the glass and is absorbed by material inside it which re-radiates it as long waves, which are trapped inside by the glass. This is called Green House effect.



- Black Surface: has highest absorption rate and highest emission coefficient for all wavelength (λ) of light.

→ Wien's law :-

* $\boxed{\lambda_{max} T = b}$ where λ_{max} = wavelength of maximum radiation (m).

* $\boxed{\lambda_{max} = \frac{2.898 \times 10^{-3}\ mK}{T\ (K)}}$ T = absolute temperature (K)

b = Wien's displacement constant ($2.898 \times 10^{-3}\ m \cdot K$)

Solar energy into heat

Green house effect → short waves (wavelength)

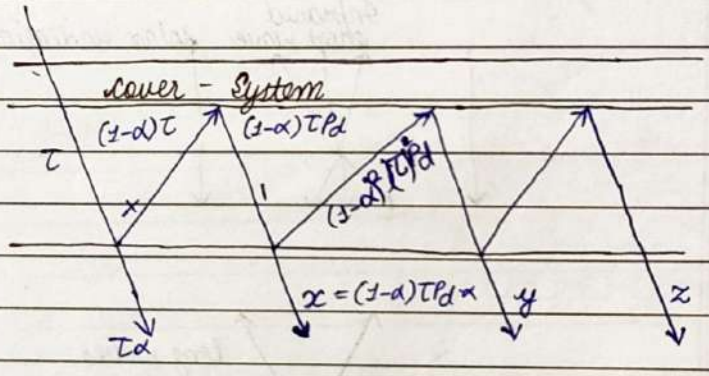
- Transmittance-Absorptance product ($\tau\rho$)

$\tau\rho$ ⇓ reflection

$\tau\alpha$ ⇓ absorption

Total transmissitivity, $\tau = \tau_p \tau_\alpha$

Definition :- This transmission and absorption product is defined as the ratio of the radiation absorbed in the absorber plate to the radiation incident on the cover system.



where, ρ_d = reflection of the cover plate for diffused reflection.

$\tau\rho_d - \alpha\tau\rho_d - \tau + \alpha\tau$

$\langle\tau\alpha\rangle = \tau\alpha + ?$

$\langle\tau\alpha\rangle = \tau\alpha + x + y + z + \ldots$

$\tau, \tau\alpha, (1-\alpha)\tau, (1-\alpha)\tau\rho_d, \ldots$

$(\alpha-1)\tau$ $\quad (1-\alpha)\tau - \tau\alpha$

$(1-2\alpha)\tau$

$$\star \quad (\tau\alpha) = \frac{\tau\alpha}{1-(1-\alpha)\rho_d}$$

Drive the complete equation for transmittance - absorptance rate product.

(8 Marks) ★★★

$$\langle \tau\alpha \rangle = \tau\alpha + \tau\alpha(1-\alpha)\rho_d + \tau\alpha(1-\alpha)^2\rho_d^2 + \tau\alpha(1-\alpha)^2\rho_d^3 + \ldots$$

$$= \tau\alpha\left[1 + (1-\alpha)\rho_d + (1-\alpha)^2\rho_d^2 + (1-\alpha)^3\rho_d^3 + \ldots\right]$$

$$= \tau\alpha \frac{1}{1-(1-\alpha)\rho_d} = \frac{\tau\alpha}{1-(1-\alpha)\rho_d}$$

Due to reflection $\langle \tau\alpha \rangle$ would be greater than $(\tau\alpha)$.

Energy Balance Equation

Under steady-state condition :-

★ Useful heat delivered by the solar collector = (Solar energy absorbed by the metal surface) − (heat losses from the surface of the surroundings)

★ $Q_u = A_c\left[HR(\tau\alpha)_e - U_L(t_p - t_a)\right]$

where Q_u = useful energy delivered by the collector (watts)

A_c = collector area (m^2)

HR = Solar energy radiation received on upper surface of collector surface.

H = rate of incident beam or diffused radiation on unit area of upper surface.

R = factor to convert beam or diffused radiation on upper surface into that on collector plane.

τ = Transmissitivity

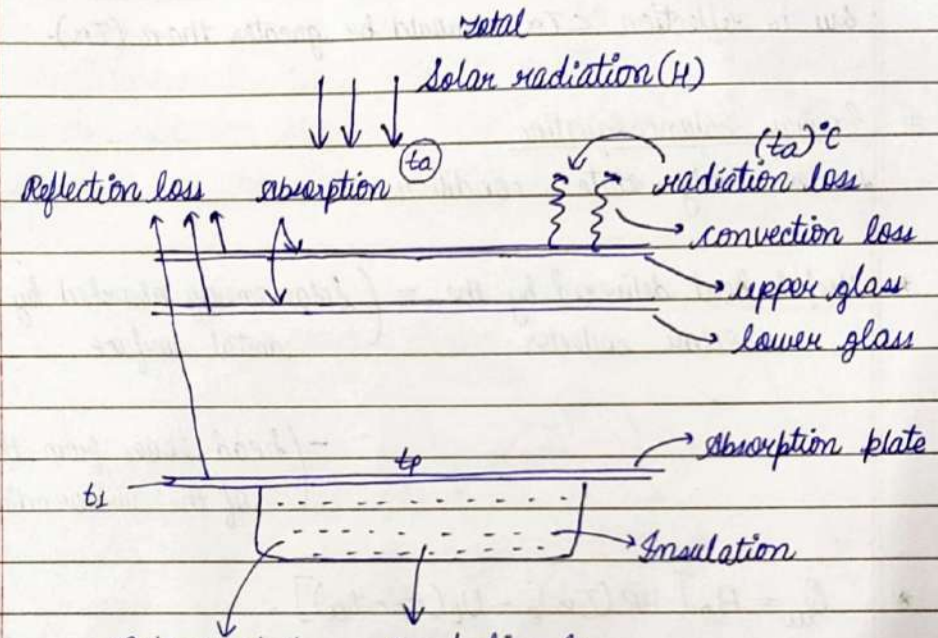
α = Absorptivity

$(\tau\alpha)_e$ = Effective transmittance absorptance product of cover surface for beam and diffused radiation

U_L = overall heat loss coefficient (W/m²°C or Kcal/m²h°C)

t_p = Average temperature of upper surface or absorber plate (°C)

t_a = Atomspheric temperature (°C)



- Absorbed energy $= A_c HR \cdot \tau \cdot \alpha$
- Effective heat loss $= A_c \cdot U_L \cdot (t_p - t_a)$.

Energy balance, $Q_u = Q_{absorbed} - Q_{loss}$

$Q_u = A_c HR \cdot \tau \cdot \alpha - A_c \cdot U_L \cdot (t_p - t_a)$

* $\boxed{Q_u = A_c [HR \cdot \tau \cdot \alpha - U_L (t_p - t_a)]}$

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Unit-I Recap

(I) Solar cell (I-V, P-V)
(II) Solar thermal energy
(III) Solar energy collector
(IV) Application
(V) Solar thermal plants

Energy balance equation of solar collector

$$\star \boxed{Q_u = A_c[H_R(\tau\alpha)_e - U_L(t_p - t_a)]}$$

How to increase energy efficiency? (Solar collectors).

a) Absorption rate $H_R(\tau\alpha)_e$ must be higher.
b) Losses $(t_p - t_a)$ must be minimum.
c) High quality transparent glass.
d) A second transparent cover with air between them.
e) Reducing radiation emitting characteristics.

Solar collector efficiency

It is a measure of collector performance and is defined as the ratio of useful gain over any time period to the incident solar energy over the same time period.

$$\star \boxed{\eta = \frac{\int Q_u dT}{\int H_R dT}}$$

Solar energy storage

→ Supply of solar energy varies with time.
→ Demand of solar energy varies with time.
→ Demand time varies with supply time.
→ Need of storage: so that it may meet the demand.

Solution :-

- Thermal energy storage
  - water storage
  - Pebble-Bed storage
- Electrical energy storage
  - Capacitor
  - Inductor
  - Battery
- Mechanical
  - chemical energy storage
  - Electromagnetic energy storage
  - ~~Water Energy~~

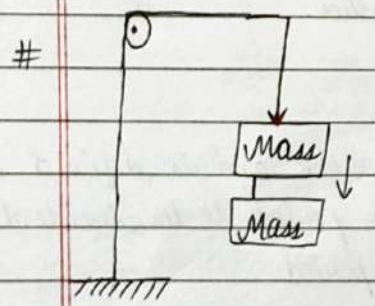
chemical energy storage

- chemical
- Thermo-chemical
- Hydrogen storage

Mech
- Pumped Hydro electric (NHPC)
- compressed air
- Flywheel

NOTE:-

$\sqrt{2} = 1.414$



Solar Energy storage

① Thermal storage

- the energy can be stored by heating and melting a materials.
- When process is reversed, the energy becomes available as heat.
- storage causing a material to rise in temperature.

The basic equation for an energy storage unit operating over finite temperature difference is

$$\star \boxed{Q_s = (mC_p)_s (T_1 - T_2)}$$

where, Q_s = Energy storage capacity
T_1, T_2 = Temperature
m = weight in kg
C_p = Specific heat

- Modification of equation :-

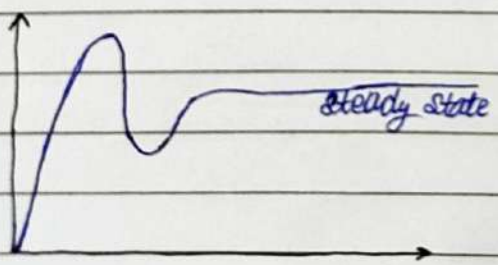
$$Q = \alpha \Delta T + \beta \Delta T^2 + \gamma \Delta T^3$$
$$= \alpha (T_1 - T_2) + \beta (T_1 - T_2)^2 + \gamma (T_1 - T_2)^3 \longrightarrow (1)$$

$$\star \boxed{Q = \alpha e^{\Delta T}}$$

→ for container of volume (V) :-

$$\boxed{\frac{Q}{V} = \rho C_p \Delta T}$$ where ρ is density of storage medium
C_p is high for water, Glauber's salt ($Na_2SO_4 \cdot 10H_2O$)

$$\star \boxed{Q = \frac{mC_{p0}}{\beta} (e^{\beta T_2} - e^{\beta T_1})}$$ where m = mass of material
C_{p0} = reference specific heat.
β = temperature coefficient
T_1, T_2 = initial & final temp.



$$\frac{dT}{dt} = -\frac{hA}{mC_p}\left(T(t) - T(\infty)\right) \longrightarrow (I)$$

$$T(t) = T(\infty) + (T_i - T_\infty)\, e^{-\left(\frac{hA}{mC_p}\right)t} \longrightarrow (II)$$

$$Q(t) = mC_p(T_i - T_\infty)$$

$$Q(t) = mC_p(T_i - T_\infty)\, e^{-\left(\frac{hA}{mC_p}\right)t} \longrightarrow (III)$$

Electrical Storage

$$\text{Heat} = \frac{1}{2} V \epsilon E^2$$

★ $$\text{Efficiency, } \eta = \frac{\int_0^{t_2} I_1 E_1 \, dT}{\int_0^{t_2} I_2 E_2 \, dt}$$

★ $$\dot{x} = -x, \quad x(0) = 1$$

I_1, I_2 = Battery discharge current

E_1, E_2 = Battery discharge terminal voltage

$$\frac{dT}{dt} = -\frac{hA}{mC_p}\left(T(t) - T(\infty)\right) \longrightarrow (I)$$

$$T(t) = T(\infty) + (T_i - T\infty)\, e^{-\left(\frac{hA}{mC_p}\right)t} \longrightarrow (II)$$

$$Q(t) = mC_p(T_i - T\infty)$$

$$Q(t) = mC_p(T_i - T\infty)\, e^{-\left(\frac{hA}{mC_p}\right)t} \longrightarrow (III)$$

Electrical Storage

$$\boxed{\text{Heat} = \frac{1}{2} V \epsilon E^2}$$

★ $$\boxed{\text{Efficiency, } \eta = \frac{\int_0^{t_1} I_1 E_1 \, dT}{\int_0^{t_2} I_2 E_2 \, dt}}$$

★ $$\boxed{\begin{aligned} \dot{x} &= -x \\ x(0) &= 1 \end{aligned}}$$

I_1, I_2 = Battery discharge current
E_1, E_2 = Battery discharge terminal voltage

Recap

① Thermal Storage:-

★ $$\boxed{Q = mC_p(T_1 - T_2)}$$

★ $$\boxed{\frac{Q}{V} = \rho C_p \Delta T}$$

② Electrical Storage:-

★ $$\boxed{H_{cap} = \frac{1}{2} V \epsilon E^2}$$

- Energy stored in capacitor is given by:-

$$\star \boxed{U_{cap} = \frac{1}{2} V \epsilon E^2}$$

where V = Volume of dielectric
E = Electric field strength
ε = permittivity of material.

Electrical Storage

- The electrical field strength is limited by breakdown strength (E) of the dielectric and therefore the electrical energy storable is also limited.

- Mica is generally ~~the~~ used dielectric material.

~~Energy stored in~~

- While the capacitors store energy at high voltage and low current, the inductors store energy at low voltage and high current.

- $\boxed{U_{inductance} = \frac{1}{2} V \mu H_m^2}$ where H_m = magnetic flux density
μ = ~~transmission~~ permeability of the material.

Solar Cell ~~photovoltaic~~

Works via ~~photoelectric~~ effect, convert sunlight into direct current (DC). when photons strike semiconductor materials, they ~~knock~~ electrons free. An electric field at the p-n junction separates these charges and metal contacts collect them to power an external circuit.

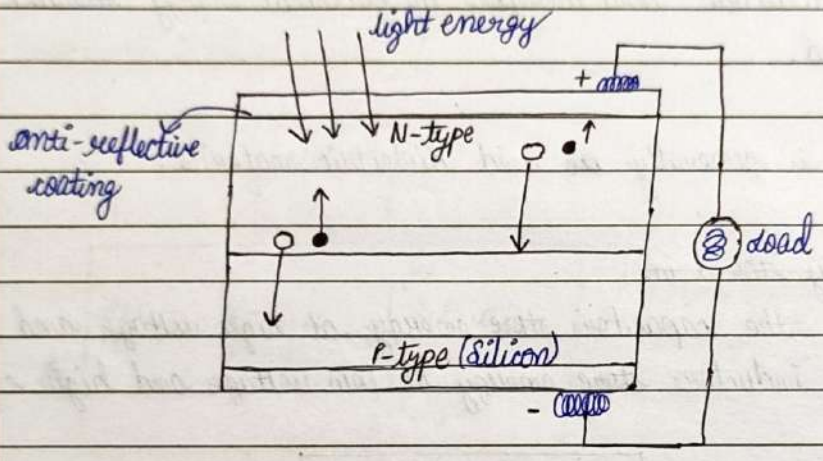
Key components:-

a) p-type:- layer doped with an elements like Boron (B) to create a positive side with missing electrons.

b) N-type :- layed doped with phosphorous (P) to provide extra free electrons.

c) P-N junction :- The boundary between the two layers, forms the electric field.

d) ~~Metal~~ Metal contacts :- conductive grids on the top and bottom that collect the lead electrons.



o Emissing Electrons - holes
• Electrons

Wind Energy

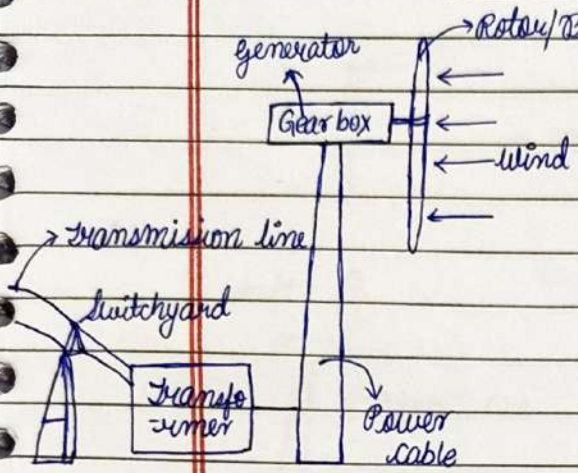
what is wind?

Wind is the horizontal movement of air from region of high pressure to a region of low pressure.

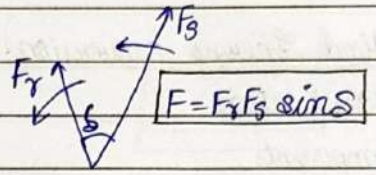
Power, $P \propto V^n$ where $n=3$ — (Governing equation).

* $P = \frac{1}{2} \rho A V^3$ where V = wind velocity

$P_{wind} \propto v^3$ where v = wind velocity



How wind turbine works?



★ $F_S \propto N_i$ where N_i = number of blades

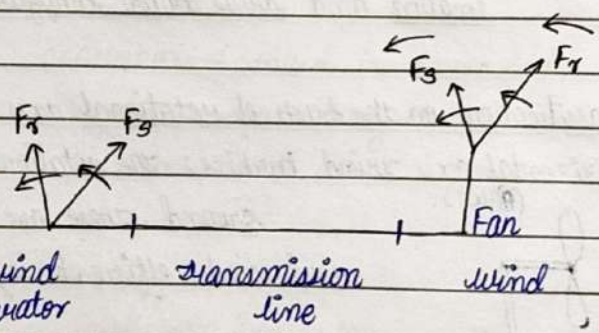
- Transformer → Voltage (up ↑ or down ↓)
- Ampere-turn relation for an ideal transformer:-

$N_1 I_1 = N_2 I_2$ where n = no of turns in coil

I = current in coil

(1 → Primary coil, 2 → Secondary coil).

- A wind energy conversion system = captures the kinetic energy of moving air and converts it into usable electrical energy.



F_R = MMF rotor

F_S = MMF Stator

★ (Mechanical energy → Electrical energy).

Electrical energy
Mechanical

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Steps of generating wind energy

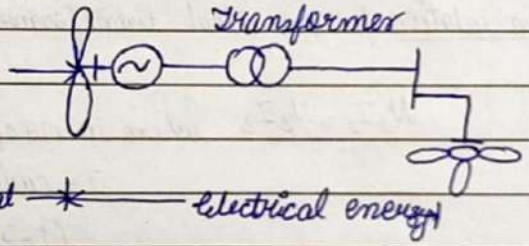
→ Wind Formation
→ Aerodynamics capture
→ Mechanical transmission
→ Electricity Generation
→ Grid Integration

Wind Energy conversion

A) Classification
B) components
c) Wind Energy Schemes

★ $P = v^3$

where v = wind speed

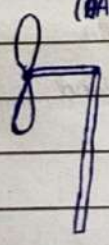


Wind Energy conversion system (WECS)

A) Classification : WECS is classified based on its rotational axis, operational location and drive train configuration.

Classification on the basis of rotational axis

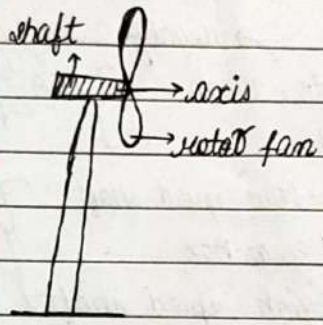
(i) Horizontal axis wind turbine: The rotational shaft is parallel to the ground. These are the most commonly used and efficient system in wind farm. (HAWT).

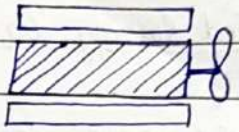


(ii) Vertical axis wind turbine :- the rotating shaft is perpendicular to the ground. These accept wind from any direction without needing yaw mechanism.

* very less efficient







Rotational axis

Speed
- → fixed speed (the rotor returns at nearly constant speed tied directly to the grid frequency)
  → simple, reliable but less efficient.
- → Variable speed → the rotor changes to match the wind speed using power electronics converters
  → (High efficiency but complex).

Location
- → OFFSHORE :- Installed in large bodies of water like oceans. They capture stronger wind, complex to install and costly.
- → ONSHORE :- Installed on land. They are easier to build and maintain, but they face land use limits and lower avg wind speed.

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Wind Energy

Wind power is proportional to third power of the wind speed.

$$P = \frac{1}{2} \rho A v^3$$ where A = area, v = wind speed

Wind Energy conversion

1. Aerodynamic components → Rotor Blade (→ Usually 2 or 3 aerodynamic blades that captures wind energy) → Hub } CFD
2. Mechanical conversion components → low speed shaft, → gear Box, → High speed shaft } Mechanical energy
3. Electrical components → Generator (Gnd), → Power converter, → Transformers } Electric Energy
4. Control and safety system
   - Controller
   - Yaw system
   - Pitch system

   → Electric + ECE

1* and cause the rotor to spin.

Hub: The central solid piece that connects the rotor blades to the main shaft.

2. Low speed shaft: connects directly to the rotor, hub and turns at a slow speed when the wind moves the blades.

Gear Box:- Increases the slow rotational speed to the low-speed shaft to high speed shaft.

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High Speed shaft :-

Spins at high speed driving the generator to produce electricity.

